

```
from docx import Document
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from matplotlib.lines import Line2D
import numpy as np
import re
```

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# -----
DOCX_PATH = "Vacinas_ingles.docx"
OUTPUT_PNG = "forest_status_vacinal.png"
# -----
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# 1. LEITURA DO DOCX
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def read_word_table(file_path, table_index=0):
    doc = Document(file_path)
    table = doc.tables[table_index]
    # Esta tabela tem 3 linhas de cabeçalho; os dados começam na linha 3
    data = []
    for row in table.rows[3:]:
        data.append([cell.text.strip() for cell in row.cells])
    return pd.DataFrame(data, columns=["label", "n_geral", "n_obito",
                                      "OR_IC", "p_OR", "ORa_IC", "p_ORa"])
```

```
def parse_or_ci(text):
    """
    Parseia strings do tipo '1.21 (0.78-1.86)'.
    Aceita vírgula ou ponto como decimal, e - ou - como separador do IC.
    Retorna (OR, IC_inf, IC_sup) ou (None, None, None) se inválido.
    """
    text = text.replace(",", ".").strip()
    match = re.search(r"([\d.]+)\s*(([\d.]+)\s*[\-\s]*([\d.]+))", text)
    if match:
        try:
            or_ = float(match.group(1))
            lo = float(match.group(2))
            hi = float(match.group(3))
            # descarta casos inestimáveis (OR ~ 0 ou IC infinito)
            if or_ <= 0 or lo <= 0 or hi == float("inf") or or_ < 1e-6:
                return None, None, None
            return or_, lo, hi
        except ValueError:
            return None, None, None
    return None, None, None
```

```
def parse_p(text):
    """Converte string de p-valor para float. '<0,001' vira 0.0005 (sentinela)."""
    text = text.replace(",", ".").strip()
    if text.startswith("<"):
        try:
            return float(text[1:]) / 2
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        except ValueError:
            return None
    try:
        return float(text)
    except ValueError:
        return None

# — Carregar e limpar


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df_raw = read_word_table(DOCX_PATH)

# Limpar labels com \n
df_raw["label"] = df_raw["label"].str.replace("\n", " / ", regex=False)

# Parsear OR bruto e OR ajustado
df_raw[["OR", "IC_inf", "IC_sup"]] = df_raw["OR_IC"].apply(
    lambda x: pd.Series(parse_or_ci(x)))
df_raw[["ORa", "ICa_inf", "ICa_sup"]] = df_raw["ORa_IC"].apply(
    lambda x: pd.Series(parse_or_ci(x)))

df_raw["p_OR"] = df_raw["p_OR"].apply(parse_p)
df_raw["p_ORa"] = df_raw["p_ORa"].apply(parse_p)

# Flags de referência e header
df_raw["is_ref"] = df_raw["OR_IC"].str.strip().str.lower() == "ref."
df_raw["is_header"] = (df_raw["n_geral"].str.strip() == "") & (~df_raw["is_ref"])

df_raw = df_raw.reset_index(drop=True)

print("=" * 70)
print(f"Tabela lida de: {DOCX_PATH}")
print(df_raw[["label", "n_geral", "n_obito", "OR", "IC_inf", "IC_sup",
               "ORa", "ICa_inf", "ICa_sup", "is_ref", "is_header"]].to_string())
print("=" * 70)

#


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# 2. FOREST PLOT
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# — Paleta


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BG      = "#FFFFFF"
PANEL   = "#F6F8FA"
BORDER  = "#D0D7DE"
TEXT    = "#1F2328"
SUBTEXT = "#57606A"
GOLD    = "#B08800"
COR_PROT = "#1D9E75" # OR < 1 → protetor
COR_RISCO = "#E07B39" # OR > 1 → risco

def sig_stars(p):
    if p is None: return ""
    if p < 0.001: return "****"
    if p < 0.01:  return "***"
    if p < 0.05:  return "**"

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return ""

# — Figura: 4 colunas — [labels] | [OR bruto] | [gap] | [OR ajustado] —————
n_rows = len(df_raw)
fig_h = max(10, n_rows * 0.56 + 4.0)

fig, axes = plt.subplots(
    1, 4, figsize=(20, fig_h), facecolor=BG,
    gridspec_kw={"width_ratios": [2.2, 3, 1.2, 3], "wspace": 0.15}
)
ax_labels, ax_or, ax_gap, ax_ora = axes

# Coluna de espaçamento invisível
ax_gap.set_visible(False)

# — Pannel de labels (sem eixos)


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ax_labels.set_facecolor(BG)
ax_labels.set_xlim(0, 1)
ax_labels.set_ylim(n_rows - 0.5, -0.5)
for spine in ax_labels.spines.values():
    spine.set_visible(False)
ax_labels.set_xticks([])
ax_labels.set_yticks([])

# Faixas zebreadas (sincronizadas nos 3 painéis)
for ax in (ax_labels, ax_or, ax_ora):
    stripe = 0
    for i, row in df_raw.iterrows():
        if not row["is_header"]:
            bg_col = "#E8EDF2" if stripe % 2 == 0 else PANEL
            ax.axhspan(i - 0.42, i + 0.42, color=bg_col, alpha=0.55, zorder=0)
            stripe += 1

# Labels no painel esquerdo (sem repetição no painel direito)
for i, row in df_raw.iterrows():
    lbl = row["label"]
    if row["is_header"]:
        lbl = lbl.split("/")[0].strip()
        ax_labels.text(0.98, i, lbl, color=TEXT,
                      fontsize=30, fontweight="bold",
                      va="center", ha="right")
    elif row["is_ref"]:
        ax_labels.text(0.95, i, f" {lbl}", color=SUBTEXT,
                      fontsize=20, va="center", ha="right")
    else:
        ax_labels.text(0.95, i, f" {lbl}", color=TEXT,
                      fontsize=22, va="center", ha="right")

# — Função genérica de painel OR


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def draw_panel(ax, col_or, col_lo, col_hi, col_p, title, xlim):
    ax.set_facecolor(PANEL)
    ax.set_xscale("log")
    ax.xaxis.grid(True, color=BORDER, linewidth=1.7, zorder=0, alpha=0.9)
    ax.set_axisbelow(True)
    for spine in ax.spines.values():
        spine.set_edgecolor(BORDER)

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spine.set_linewidth(0.9)

ax.axvline(1.0, color=GOLD, linewidth=2.3, linestyle="--", zorder=2, alpha=0.9)

for i, row in df_raw.iterrows():
    if row["is_header"]:
        continue

    if row["is_ref"]:
        ax.text(0.5, i, "ref.", color=SUBTEXT, fontsize=20,
                va="center", ha="center",
                transform=ax.get_yaxis_transform())
        continue

    OR = row[col_or]
    lo = row[col_lo]
    hi = row[col_hi]
    p = row[col_p]

    # OR inestimável
    if pd.isna(OR) or pd.isna(lo) or pd.isna(hi):
        ax.text(0.5, i, "—", color=SUBTEXT, fontsize=20,
                va="center", ha="center",
                transform=ax.get_yaxis_transform())
        continue

    cor = COR_PROT if OR < 1 else COR_RISCO
    sig = (p is not None) and not np.isnan(p) and (p < 0.05)

    # Barra IC capada nos limites do eixo
    lo_plot = max(lo, xlim[0] * 1.02)
    hi_plot = min(hi, xlim[1] * 0.98)
    ax.plot([lo_plot, hi_plot], [i, i],
            color=cor, linewidth=3.8, zorder=3, alpha=0.85,
            solid_capstyle="round")

    # Marcador: diamante se significativo, círculo se não
    ax.plot(OR, i,
            marker="D" if sig else "o",
            markersize=8 if sig else 7,
            color=cor,
            markerfacecolor=cor if sig else BG,
            markeredgecolor=cor,
            markeredgewidth=1.6,
            zorder=5)

    # Texto OR (IC) + estrelas à direita do painel
    hi_txt = min(hi, 9999)
    txt = f"OR:.2f ( {lo:.2f}–{hi_txt:.2f} ) {sig_stars(p)}"
    ax.text(1.02, i, txt,
            transform=ax.get_yaxis_transform(),
            color=TEXT, fontsize=20, va="center", ha="left",
            clip_on=False)

ax.set_yticks(range(n_rows))
ax.set_yticklabels([""] * n_rows)
ax.tick_params(axis="y", length=0)
ax.tick_params(axis="x", colors=SUBTEXT, labels=12)
ax.set_ylim(n_rows - 0.5, -0.5)
ax.set_xlim(*xlim)

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ax.set_xlabel("Odds Ratio (scale log)", fontsize=25,
              color=SUBTEXT, labelpad=8)
ax.set_title(title, fontsize=30, fontweight="bold", color=TEXT, pad=12)

draw_panel(ax_or, "OR", "IC_inf", "IC_sup", "p_OR",
           "Crude OR (95% CI)", xlim=(0.08, 32))
draw_panel(ax_ora, "ORa", "ICa_inf", "ICa_sup", "p_ORa",
           "Adjusted OR (95% CI)", xlim=(0.08, 32))

# — Legenda


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legend_elements = [
    mpatches.Patch(facecolor=COR_PROT, edgecolor=COR_PROT,
                   label="Protective factor (OR < 1)"),
    mpatches.Patch(facecolor=COR_RISCO, edgecolor=COR_RISCO,
                   label="Risk factor (OR > 1)"),
    Line2D([0], [0], marker="D", color="none",
           markerfacecolor=TEXT, markeredgecolor=TEXT,
           markersize=5, label="Diamond = p < 0,05"),
    Line2D([0], [0], marker="o", color="none",
           markerfacecolor=BG, markeredgecolor=TEXT,
           markeredgewidth=1.4, markersize=7,
           label="Open circle = p ≥ 0,05"),
    Line2D([0], [0], color=GOLD, linewidth=1.5,
           linestyle="--", label="Reference line (OR = 1)"),
]
ax_or.legend(handles=legend_elements, fontsize=10, frameon=True,
             edgecolor=BORDER, facecolor=BG, labelcolor=TEXT,
             loc="lower right", framealpha=0.97,
             borderpad=0.9, handlelength=0.5)

# — Título geral


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fig.text(0.5, 1.025,
        "Crude and adjusted odds ratios",
        ha="center", va="top",
        fontsize=35, fontweight="bold", color=TEXT)
fig.text(0.5, 0.978,
        "Vaccination status | n = 703",
        ha="center", va="top", fontsize=25, color=SUBTEXT)

fig.add_artist(plt.Line2D(
    [0.03, 0.97], [0.938, 0.938],
    transform=fig.transFigure, color=BORDER, linewidth=1.8))

fig.text(0.03, 0.003,
        "**** p<0,001  ** p<0,01  * p<0,05 | "
        "OR = Odds Ratio; CI = 95% Confidence Interval | "
        "Filled diamond = p < 0,05 | — = Non-estimable OR",
        color=SUBTEXT, fontsize=18, style="italic")

plt.tight_layout(rect=[0, 0.018, 1, 0.970])
plt.savefig(OUTPUT_PNG, dpi=180, bbox_inches="tight", facecolor=BG)
print(f"\n Gráfico salvo em: {OUTPUT_PNG}")
plt.show()

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